Free balancing equations worksheet worksheet 15

Maths | 9th grade | Algebra

1. If 3(x - 2) = 2(x + 5) + 4, what is the value of x?

A) 4
B) 6
C) 8
D) 10

2. Solve for y: 4y + 12 = 3(y - 4) + y

A) -12
B) 0
C) 12
D) 4

3. Find the solution to the equation: 5(x + 3) = 3x + 18

A) 1
B) 2
C) 3
D) 4

4. What is the value of k if 2k - (3 - k) = k + 5?

A) 1
B) 2
C) 3
D) 4

5. If 7x - 4 = 2x + 16, what is x?

A) 4
B) 6
C) 8
D) 10

6. Solve for n: 3n + 2 = 2n + 5

A) 1
B) 2
C) 3
D) 4

7. If 4(x + 1) = x + 13, what is the value of x?

A) 3
B) 2
C) 4
D) 5

8. What is the solution to 2(m - 3) = 4 - m?

A) -1
B) 1

C) 3
D) 5

9. Solve for p: p/2 + 3 = 7

A) 4
B) 6
C) 8
D) 10

10. If 5q - 9 = 2(q + 4), what is the value of q?

A) 2
B) 3
C) 4
D) 5

Answer Key

1. B
Distribute and simplify both sides: 3x - 6 = 2x + 10 + 4, then solve for x: x = 6.

2. B
Distribute and combine like terms: 4y + 12 = 3y - 12 + y, then solve for y: y = 0.

3. C
Distribute: 5x + 15 = 3x + 18, then solve for x: x = 3.

4. B
Simplify: 2k + k - 3 = k + 5, then solve for k: k = 2.

5. A
Subtract 2x from both sides and add 4 to both sides: 5x = 20, so x = 4.

6. D
Subtract 2n from both sides and subtract 2: n = 3.

7. A
Distribute and solve: 4x + 4 = x + 13, subtract x and 4: 3x = 9, x = 3.

8. B
Expand and solve: 2m - 6 = 4 - m, add m and 6: 3m = 10, m = 1.

9. C
Subtract 3 from both sides and multiply by 2: p/2 = 4, p = 8.

10. B
Distribute and solve: 5q - 9 = 2q + 8, subtract 2q and add 9: 3q = 17, q = 3.